



Advanced Linear Algebra: Foundations to Frontiers

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2.2.2 Component in the direction of a vector

02.2.2 Component in the direction of ...

fit width



In a previous linear algebra course, you may have learned that if $a, b \in \mathbb{R}^m$ then

$$\hat{b} = \frac{a^T b}{a^T a} a = \frac{a a^T}{a^T a} b$$

equals the component of b in the direction of a and

$$b^\perp = b - \hat{b} = \left(I - \frac{a a^T}{a^T a}\right) b$$

equals the component of b orthogonal to a , since $b = \hat{b} + b^\perp$ and $\hat{b}^T b^\perp = 0$.
Similarly, if $a, b \in \mathbb{C}^m$ then

$$\hat{b} = \frac{a^H b}{a^H a} a = \frac{a a^H}{a^H a} b$$

equals the component of b in the direction of a and

$$b^\perp = b - \hat{b} = \left(I - \frac{a a^H}{a^H a}\right) b$$

equals the component of b orthogonal to a .

Remark 2.2.2.1. The matrix that (orthogonally) projects the vector to which it is applied onto the vector a is given by

$$\frac{aa^H}{a^Ha}$$

while

$$I - \frac{aa^H}{a^Ha}$$

is the matrix that (orthogonally) projects the vector to which it is applied onto the space orthogonal to the vector a .

Homework 2.2.2.1. Let $a \in \mathbb{C}^m$.

ALWAYS/SOMETIMES/NEVER>:

$$\left(\frac{aa^H}{a^Ha} \right) \left(\frac{aa^H}{a^Ha} \right) = \frac{aa^H}{a^Ha}.$$

Interpret what this means about a matrix that projects onto a vector.

▼ Answer ▼ Solution

$$\begin{aligned} & \left(\frac{aa^H}{a^Ha} \right) \left(\frac{aa^H}{a^Ha} \right) \\ &= < \text{multiply numerators and denominators} > \\ & \frac{aa^Haa^H}{(a^Ha)(a^Ha)} \\ &= < \text{associativity} > \\ & \frac{a(a^Ha)a^H}{(a^Ha)(a^Ha)} \\ &= < a^Ha \text{ is a scalar and hence commutes to front} > \\ & \frac{a^Haaa^H}{(a^Ha)(a^Ha)} \\ &= < \text{scalar division} > \\ & \frac{aa^H}{a^Ha}. \end{aligned}$$

Interpretation: orthogonally projecting the orthogonal projection of a vector yields the orthogonal projection of the vector.

ALWAYS.

Now prove it.

Homework 2.2.2.2. Let $a \in \mathbb{C}^m$.

ALWAYS/SOMETIMES/NEVER:

$$\left(\frac{aa^H}{a^H a} \right) \left(I - \frac{aa^H}{a^H a} \right) = 0$$

(the zero matrix). Interpret what this means.

▼ Answer

▼ Solution

$$\begin{aligned} & \left(\frac{aa^H}{a^H a} \right) \left(I - \frac{aa^H}{a^H a} \right) \\ &= < \text{distribute} > \\ & \left(\frac{aa^H}{a^H a} \right) - \left(\frac{aa^H}{a^H a} \right) \left(\frac{aa^H}{a^H a} \right) \\ &= < \text{last homework} > \\ & \left(\frac{aa^H}{a^H a} \right) - \left(\frac{aa^H}{a^H a} \right) \\ &= \\ & 0. \end{aligned}$$

Interpretation: first orthogonally projecting onto the space *orthogonal to* vector a and then orthogonally projecting the resulting vector onto that a leaves you with the zero vector.

ALWAYS.

Now prove it.

Homework 2.2.2.3. Let $a, b \in \mathbb{C}^n$, $\hat{b} = \frac{aa^H}{a^H a} b$, and $b^\perp = b - \hat{b}$.

ALWAYS/SOMETIMES/NEVER: $\hat{b}^H b^\perp = 0$.

▼ Answer

▼ Solution

$$\hat{b}^H b^\perp$$

$$= \langle \text{substitute } \hat{b} \text{ and } b^\perp \rangle$$

$$\left(\frac{aa^H}{a^H a} b \right)^H (b - \hat{b})$$

$$= \langle (Ax)^H = x^H A^H; \text{ substitute } b - \hat{b} \rangle$$

$$b^H \left(\frac{aa^H}{a^H a} \right)^H \left(I - \frac{aa^H}{a^H a} \right) b$$

$$= \langle ((xy^H)/\alpha)^H = yx^H/\alpha \text{ if } \alpha \text{ is real} \rangle$$

$$b^H \frac{aa^H}{a^H a} \left(I - \frac{aa^H}{a^H a} \right) b$$

$$= \langle \text{last homework} \rangle$$

$$b^H 0 b$$

$$= \langle 0x = 0; y^H 0 = 0 \rangle$$

$$0.$$

ALWAYS.

Now prove it.